

# Antibody drug conjugate (ADC) treatment

Antibody drug conjugates (ADCs) are a type of cancer drug treatment that links a targeted therapy with a chemotherapy drug. ADCs are sometimes called smart chemotherapy.

## On this page

- [What are antibody drug conjugates \(ADCs\)?](#)
- [Antibody drug conjugate treatments](#)
- [Your course of antibody drug conjugate treatment](#)
- [How antibody drug conjugates are given](#)
- [About side effects of antibody drug conjugate treatment](#)
- [Main side effects of antibody drug conjugate treatment](#)
- [More information about antibody drug conjugates](#)
- [Get in touch](#)

## What are antibody drug conjugates (ADCs)?

Antibody drug conjugates (ADCs) are a type of cancer drug treatment. They are mainly used to treat cancers such as [lymphoma](#), [leukaemia](#) and [breast cancer](#). They are sometimes used to treat other types of cancer, including [bladder cancer](#) and [upper urinary tract urothelial cancer \(UTUC\)](#).

ADCs link together a [targeted therapy](#) called a monoclonal antibody with a chemotherapy drug. The chemical that links them together is called a linker. The combination of monoclonal antibody, chemotherapy and linker is called a conjugate.

ADCs work by delivering chemotherapy directly to cancer cells while minimising damage to healthy cells.

The antibody is designed to target a specific protein. This protein is found mostly on the surface of the cancer cells, and much less on the healthy cells. The antibody attaches to the specific protein.

Once attached to the protein on the surface of the cancer cell, the antibody and the chemotherapy drug it is linked to get absorbed. The chemotherapy can then work to destroy the cancer cell.

Clinical trials have shown that some ADCs may also destroy nearby cancer cells even if they do not have the specific protein. This is called the bystander effect.

*Image: The diagram shows how an antibody drug conjugate (ADC) works on a cancer cell in 3 parts. The first part of the diagram shows a close-up of a Y-shaped monoclonal antibody. One arm of the Y is attached to the surface of a cancer cell. A short black line labelled "Linker" connects the base of the antibody to a small red circle labelled "Chemotherapy". The second part of the diagram shows the full the cancer cell. Several small Y-shaped ADCs are attached to its surface. A small circle shows an ADC being taken into the cancer cell. An arrow leads to the next circle inside the cancer cell, where the antibody breaks apart and the chemotherapy separates from it. An arrow points to the final circle, where the chemotherapy is released into the cancer cell. A large arrow points down from the cancer cell to the third part of the diagram. This shows an irregular, damaged shape. This is the cancer cell that has been destroyed because of ADC treatment.*

Because ADCs specifically target cancer cells and not the healthy cells, ADC treatment can help reduce chemotherapy side effects.

## Antibody drug conjugate treatments

Your cancer team can talk to you about ADC treatment and whether it is suitable for you. There are several ADCs available for different cancer types. This includes:

- [gemtuzumab ozogamicin \(Mylotarg<sup>®</sup>\)](#).
- [inotuzumab ozogamicin \(Besponsa<sup>®</sup>\)](#).
- [brentuximab vedotin \(Adcetris<sup>®</sup>\)](#).
- polatuzumab vedotin (Polivy<sup>®</sup>)
- [loncastuximab tesirine \(Zynlonta<sup>®</sup>\)](#).
- belantamab mafodotin (Blenrep<sup>®</sup>)
- enfortumab vedotin (Padcev<sup>®</sup>)
- [trastuzumab emtansine \(Kadcyla<sup>®</sup>\)](#).
- [trastuzumab deruxtecan \(Enhertu<sup>®</sup>\)](#).
- [sacituzumab govitecan \(Trodelvy<sup>®</sup>\)](#).
- tisotumab vedotin
- [mirvetuximab soravtansine \(ELAHERE<sup>®</sup>\)](#).

ADCs can also be given in combination with other types of cancer treatments, such as [chemotherapy](#) or [immunotherapy](#). For example, this may include:

- [gemtuzumab ozogamicin \(Mylotarg<sup>®</sup>\) with daunorubicin and cytarabine \(DA\)](#).
- [pola-BR](#)
- [pola-R-CHP](#)
- [brentuximab vedotin \(Adcetris<sup>®</sup>\) with doxorubicin, vinblastine and dacarbazine \(BV-AVD\)](#).
- [belantamab mafodotin, bortezomib and dexamethasone \(BVD\)](#).
- [enfortumab vedotin \(Padcev<sup>®</sup>\) with pembrolizumab \(Keytruda<sup>®</sup>\)](#).

If you know the name of the drug or combination, you can use our list of treatments to find it. This gives more information about:

- how you have the treatment
- possible side effects.

## Your course of antibody drug conjugate treatment

Your cancer team will discuss your ADC treatment plan and possible side effects with you before you agree ([consent](#)) to have treatment.

How often you have treatment, how it is given and how long your course of treatment lasts depends on:

- the type of cancer
- the ADC you are having
- how the cancer responds to the treatment
- any side effects you have.

ADCs are often given as several sessions of treatment. You have rest periods in between each session of treatment. You have no treatment during the rest period. The rest period allows your body to recover from the side effects. The treatment session and the rest period make up one cycle of treatment. Your cancer team can explain how many cycles you may need.

## How antibody drug conjugates are given

Antibody drug conjugates (ADCs) are given as a drip into a vein. This is called an intravenous infusion. Your cancer team can explain what to expect.

Depending on the type of cancer, you will either have treatment:

- at a day unit or clinic as an outpatient
- during a stay in hospital.

The nurse will explain how long your treatment will take.

You may have treatment into a vein through:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in the arm or hand
- a [central line](#) – a fine tube that goes under the skin of the chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in the arm and goes up into a vein in the chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on the chest or arm and goes into a vein in the chest.

# About side effects of antibody drug conjugate treatment

Your cancer team will give you information about possible side effects before you start treatment. You may get some of the side effects but you are unlikely to get all of them.

You may be having an ADC in combination with other cancer treatments. If you are having other cancer treatments, you may have other side effects.

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your cancer doctor, specialist nurse or pharmacist can explain the risk of these side effects to you.

## Contacting the hospital

Your cancer team will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

# Main side effects of antibody drug conjugate treatment

Different ADC treatments can have different side effects.

To learn more about your specific treatment, you can search for information about the ADC or combination of drugs using our [cancer treatments and drugs A to Z](#).

You may get the following side effects.

## Allergic reaction

Some people have an allergic reaction while having ADC treatment. The first infusion is the most likely to cause a reaction, so it is usually given more slowly than later treatments.

Before treatment, you will have medicines to help prevent or reduce any reaction.

Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your infusion. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

**Always call 999 if swelling happens suddenly or you are struggling to breathe.**

## Risk of infection

ADC treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible.

If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection can include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

**It is important to follow any specific advice your cancer treatment team gives you.**

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

## Bruising and bleeding

ADC treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
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blood in your urine (pee) or stools (poo)

- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

## Anaemia (low number of red blood cells)

ADC treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia.

You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

## Feeling tired (fatigue)

[Feeling tired](#) is a common side effect of ADC treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

## Feeling sick

ADC treatment may cause sickness. Your cancer team will prescribe anti-sickness drugs to help prevent or control sickness.

Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

[If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

## Constipation

ADC treatment may cause constipation. Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (8 glasses) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
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- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

## Diarrhoea

ADC treatment may cause diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

## Sore mouth and throat

ADC treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection.

Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

## Skin changes

ADC treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy.

Always tell your cancer team about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

## Numb or tingling hands or feet (peripheral neuropathy)

ADC treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your cancer team. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long term side effect of treatment.

## Effects on the heart

ADC treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your cancer doctor may give you tablets to manage these. Or they may change the type of cancer treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

## Effects on the lungs

ADC treatment can cause inflammation or changes to the lungs. Contact the hospital straight away on the 24-hour number if you develop:

- a cough that does not go away
- wheezing
- breathlessness
- a fever, with a temperature over 37.5°C.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

## Effects on the liver

Some ADC treatments can affect how the liver works. You will have regular blood tests to check this.

Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen)
- swelling in the tummy (abdomen) caused by a build-up of fluid (ascites).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

## Tumour lysis syndrome (TLS)

Some people being treated for types of blood cancer such as lymphoma or leukaemia may be at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment.

TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (8 glasses) of fluid a day will also help.

## More information about antibody drug conjugates

If you need information about a specific bispecific antibody, you can check our [cancer treatments and drugs A to Z](#). We have detailed information about many types of cancer drugs and their possible side effects.

You can also talk to your cancer team if you want more detailed information about a treatment. Or visit the [electronic medicines compendium \(emc\)](#), which has patient information leaflets (PILs) for individual drugs.

## Other useful resources

If you want more information about treatment and side effects, you can visit our cancer treatment forum. Our Online Community is a place to talk with others having treatment, share your experience, and ask questions.

We also have more information about treatment in our booklets:

- [Ask about your cancer treatment](#)
- [Side effects of cancer treatment](#)

- [Making treatment decisions.](#)

## About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

### References



Below is a sample of the sources used in our antibody drug conjugates (ADCs) information. If you would like more information about the sources we use, please contact us at [informationproductionteam@macmillan.org.uk](mailto:informationproductionteam@macmillan.org.uk)

Fong JY, Phuna Z, Chong DY, Heryanto CM, Low YS, Oh KC, et al. Advancements in antibody-drug conjugates as cancer therapeutics. J Natl Cancer Cent. 2025; 5(4):362-378. Available from: [www.doi.org/10.1016/j.jncc.2025.01.007](https://www.doi.org/10.1016/j.jncc.2025.01.007) [accessed August 2025].

Pettinato MC. Introduction to antibody-drug conjugates. Antibodies (Basel). 2021 Oct 27;10(4):42. Available from: [pmc.ncbi.nlm.nih.gov/articles/PMC8628511](https://pmc.ncbi.nlm.nih.gov/articles/PMC8628511) [accessed August 2025].

Tsuchikama K, Anami Y., Ha SYY, et al. Exploring the next generation of antibody–drug conjugates. Nat Rev Clin Oncol. 2024;21:203–223. Available from: [www.doi.org/10.1038/s41571-023-00850-2](https://www.doi.org/10.1038/s41571-023-00850-2) [accessed August 2025].

Wang R, Hu B, Pan Z, Mo C, Zhao X, Liu G, et al. Antibody–drug conjugates (ADCs): current and future biopharmaceuticals. J Hematol Oncol. 2025 Apr 30;18:51. Available from: [www.doi.org/10.1186/s13045-025-01704-3](https://www.doi.org/10.1186/s13045-025-01704-3) [accessed August 2025].

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

## The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

## Get in touch

### Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

### Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



**0300 1000 200**

Our Support Line is available Monday to Friday, 9:30am-4:30pm



**Email us**

Email the Macmillan Fundraising team if you have any questions.



**Online Community**

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



## **Find cancer support services in your area**

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.